so that, as long as $\nu \leq C_2 h^{-k+r-1}$, we have

$$|H(z_\nu) - H(z_0)| \leq \mathcal{O}(h^r).$$

□

If the differential equations are infinitely differentiable, we may take the truncation index k as large as we like, but the constants appearing in the above theorem will depend on the truncation index in a complicated way. It is possible to prove (see discussions in [164, 227] for more detail), that for many standard classes of numerical methods, there are real, positive constants C, D such that

$$\|\mathcal{G}_h(\boldsymbol{\zeta}) - \mathcal{F}_h^{(k)}(\boldsymbol{\zeta})\| \leq Ch\,[D(k+1)h]^{k+1}\,,$$

giving a precise bound on the magnitude of the difference between the time h evolution under the truncated perturbed Hamiltonian and the numerical method.

If h is small, then for k sufficiently small the quantity in brackets is less than one and the difference from the truncated approximation decreases in magnitude with increasing k. As soon as k satisfies

$$k + 1 > \frac{1}{Dh}$$

the power grows monotonically without bound. We can minimize the difference between $\mathcal{G}_h$ and $\mathcal{F}_h^{(k)}$ by choosing

$$k = \frac{1}{hDe} - 1,$$

in which case,

$$\|\mathcal{G}_h(\boldsymbol{\zeta}) - \mathcal{F}_h^{(k)}(\boldsymbol{\zeta})\| < \tilde{C}he^{-\frac{1}{\gamma h}}, \qquad \gamma = \frac{1}{De}.$$

This bound tends to zero extremely rapidly (more rapidly than any power of h) as $h \to 0$. The assumption is that if this error in the approximation of the numerical map by the flow of a truncated perturbed Hamiltonian flow map is much smaller than the other errors present in our model, any drift due to lack of convergence would be minor, perhaps even invisible, on the timescale of simulation.

It is not recommended to take these estimates too literally, i.e. to try to compute the constants C and D (which might be quite difficult). Instead, we simply use the estimates to guide our intuition that the energy will be expected to fluctuate on very long time intervals with a magnitude bounded by $\mathcal{O}(h^r)$. To gain a full understanding of the issue of energy drift in the setting of molecular dynamics, the article [121] is highly relevant.